Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/15/2005
PI Name:	Roux, Stanley Ph.D.		
Project Title:	Cellular bases of light regulated gravity responses		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78712-1240	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Gregory B. (The University of Texas at Austin)		
Grant/Contract No.:	NAG2-1586		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overall goal of this project is to clarify the cellular mechanisms that control light-regulated gravity responses. Results obtained this year advanced the overall objective of this project by providing new information on steps in the transduction chain for gravity-regulated responses in which Ca2+ is likely to exert a critical influence. Here we highlight two important advances that contributed toward achieving project objectives during the current funding year. These advances are: 1. Further characterization of T-DNA insertional knockouts of each of the 8 different annexin genes in Arabidopsis. This advance allows us to use a rigorous genetic approach to evaluate the role of each of 8 different annexins in plant growth and development generally, and specifically in gravitropism. 2. Discovery that extracellular ATP (xATP) regulates growth of plant shoots and roots, and that the level of expression of a mRNA that encodes an xATP-hydrolyzing enzyme (ectoapyrase) is correlated with growth. This discovery, together with the finding that		

	increased [xATP] upregulates genes encoding enzymes that increase the production of the growth inhibitors superoxide and ethylene, and that this induction is dependent on xATP-induced increase in cytosolic calcium, suggest a mechanism to explain how application of ATP to the growth medium can inhibit the gravitropic growth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research does not directly seek to understand a disease or malady that affects humans on Earth and/or in space, nor does it seek to develop new therapeutics for alleviating symptoms of a malady on Earth. However, the findings will help to understand the growth of plants better, and since plants are a crucial source of food for humans, this research does seek to understand the malady of malnutrition. Also, this research does yield a new understanding of basic biological processes, specifically the processes of plant growth and the cellular mechanisms whereby gravity can affect the developmental polarity in cells. Further, this research points to a real role of gravity in regulating growth and development of plants on Earth and thus reveals potential problems in achieving normal growth and development of plants in space. Finally, the health of humans is inextricably linked to the ability to control and continuously improve the growth of plants. This, in turn, requires an improved understanding of the molecular mechanisms that control growth in plants. The accomplishments of this research contribute to that improved understanding, and thus indirectly benefit humans.
Task Progress:	This report summarizes the two most significant areas of research accomplished during the past year in our NAG2-1586 project on the cellular bases of light-regulated gravity responses. It elaborates mainly on our analysis of the growth effects of knocking out individual members of the annexin family of calcium-binding proteins in Arabidopsis, and on our potentially related discovery that extracellular ATP and an enzyme that regulates the turnover of extracellular ATP play a role in the regulation of plant growth and thus can be important responsive elements during the gravitropic response of roots and shoots. 1. Analysis of T-DNA insertional knockouts of Arabidopsis annexin genes. Regarding the involvement of annexins in gravity responses, last year we reported that annexin localization correlated with regions of active secretion and growth in plants. This was an important finding (manuscript on this now In press), because a key aim of the NAG2-1586 NASA project was to identify different sites of function of Ca2+-binding proteins that have been implicated in growth control, with a focus on understanding the role of these proteins in gravity responses. A prediction of this finding is that annexins are critically involved in growth processes, including gravitropism, and that mutant plants knocked out in one or more annexin genes would exhibit impaired growth. To test this prediction last year we acquired T-DNA insertional knockouts of all 8 annexin genes in Arabidopsis, as indicated in our 2003 Task Progress report. All of these lines were from the Arabidopsis Biological Resource Center and were uncharacterized when we got them. During this past year we characterized these knockout lines and, through backcrosses, have obtained single-insertion knockout lines for three of the annexin genes, and have confirmed that all three are null mutants for those annexins. In initial growth tests, we have shown that all three of these mutants are significantly shorter than wild-type plants after 7 days growth in darkness. We are currently testing whether this indicates the mutants have impaired growth or delayed germination. Either way, the mutants are clearly deficient in some growth function. Since in each case each growth-deficient mutant is missing only one annexin, this indicates that each of these three annexins serves a role in growth that is not completely complemented by the presence of other annexins. We have confirmed that the putative knockout lines for the other five annexins in Arabidopsis are also null mutants, but these all have two or more T-DNA insertions, so we are backcrossing with wild-type lines to isolate single insertion knockouts of these lines. We expect to complete most of this work in the upcoming third year of our grant. As a follow-up to these studies, we will be particularly interested to learn whether any one or combination of annexin knockouts alters the kinetics of auxin transport or of root or shoot gravitropism. 2. Extracellular ATP can regulate growth and can serve as a potential signaling agent in the transduction pathway for gravitropism. Regarding the novel discovery that extracellular ATP (xATP) and an ectoapyrase that controls the turnover of xATP control growth in plants, last year we reported that xATP could suppress gravitropic growth of roots at concentrations that did not impair overall root growth. This year we focused on the question of what signaling steps converted the xATP stimulus into a growth response. We carried out a dose-response curve plotting the elongation growth of etiolated hypocotyls vs. concentration of xATP, and we found that at very low concentrations of applied ATP growth was stimulated, but that at higher concentrations, growth was inhibited. To be sure these effects were not due to hydrolysis of ATP to generate free phosphate, a critical nutrient that would clearly influence growth, we repeated the dose-response curve using ATP-gamma S, a non-hydrolyzable analogue of ATP, and found the growth curve had the same shape as that induced by ATP, but the concentrations needed for both stimulation and inhibition of growth were 10-fold lower when the seedlings were treated with ATP-gamma S than when they were treated with ATP. The same concentrations of ATP-gamma S that induced growth inhibition also stimulated a rise in the cytosolic calcium concentration of cells in both the shoots and roots of the seedlings, and induced increased message levels for genes involved in ethylene production. We found that the ability of ATP-gamma S to induce growth inhibition was blocked by AVG, an inhibitor of ethylene synthesis, and was suppressed in ein2 mutants that were insensitive to ethylene. Ethylene treatment can disrupt normal gravitropic growth, so our data are consistent with the hypothesis that the inhibitory effect of xATP on root gravitropism may be mediated in part by ethylene. Using a different application technique we found that sub-micromolar concentrations of ATP could induce the production of superoxide, and that this effect could be blocked by agents that reduced the uptake of calcium into cells. Superoxide is rapidly converted to hydrogen peroxide, a growth regulator recently implicated in gravitropism control. Ethylene production is typically downstream of the superoxide/hydrogen peroxide signal, so it is possible that the effects of xATP on ethylene production are mediated by its earlier induction of superoxide, which, in turn, is dependent on xATP induction of increased uptake of calcium into cells, presumably through ATP-activated calcium channels. Extracellular ATP is known to activate calcium channels in animal cells, but evidence for this phenomenon in plants is just beginning to be gathered. The above data suggest that accumulation of xATP in the plant cell wall above a certain threshold would inhibit growth. Growth is typically accompanied by the delivery of secretory vesicles to the plasma membrane and by the delivery of their contents into the wall. In animals, secretory vesicles are known to contain high concentrations of ATP, and bursts of ATP are released into the extracellular matrix of cells at sites of active secretion. This process may also occur in plants. If so, then growth would be normally accompanied by an increase of [ATP] in the walls of cells. If increased [ATP] in the wall inhibits growth, then for growth to be maintained, it would be important to hydrolyze the wall ATP accumulating in growth zones. Consistent with this idea, the level of mRNA encoding the xATP-hydrolyzing ectoapyrase enzyme is highest in tissues with the highest growth rates. Stimuli that result in growth reduction (e.g., red light irradiation of etiolated seedlings) result simultaneously in reduction of apyrase mRNA levels. Taken together the results described in this section (ii) lead to the novel insight that growth control in plants involves, in part at least, control of xATP accumulation

	by the activity of ectoapyrases. They are consistent with the conclusion that the inhibition of gravitropism by xATP may be mediated in parts by signaling agents that include increased cytosolic calcium concentration, superoxide/hydrogen peroxide, and ethylene. We are currently testing the validity of this conclusion.
Bibliography Type:	Description: (Last Updated: 10/16/2023)
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